

Reyna Patel

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EDUCATION

University of Southern California

Bachelor of Science, Major in Computer Engineering and Computer Science, Minor in Mathematics

May 2028

Los Angeles, CA

Relevant Coursework: Algorithms and Theory of Computing, Principles of Software Development, Data Structures, Object Oriented Programming, Distributed Systems for the IoT, Embedded Systems, Discrete Structures, Linear Algebra and Linear Differential Equations

WORK EXPERIENCE

University of Southern California, Viterbi School of Engineering - Interaction Lab

Sep. 2025 - Present

Undergraduate Researcher

Los Angeles, CA

- Building Python ML pipelines to preprocess and model physiological heart-rate signals across hundreds of sessions
- Automating Fitbit compliance tracking and reminder emails (Python, AWS SES) for a 20-participant NIH-funded study
- Developing statistical analysis models forecasting stress trajectories for a socially assistive robot delivering CBT exercises
- Second author on 2 research manuscripts submitted to CHI 2027 and HRI 2027; contributor to manuscript submitted to IEEE T-RO

University of Southern California - Distance Education Network @ Viterbi

Aug. 2025 - Present

Camera Operator/Network Operator

Los Angeles, CA

- Operating professional broadcast equipment to record graduate-level engineering lectures for USC's Distance Education network
- Monitoring live and recorded sessions to ensure audio/visual quality, continuity, and accessibility for remote students
- Coordinating with instructors and technical staff to troubleshoot equipment and maintain smooth delivery of course content

University of Southern California - Electronic Dance & Music Association

Aug. 2025 - Sep. 2025

Software Engineering Intern

Los Angeles, CA

- Developed a mobile app using React Native for the USC Electronic Dance & Music Association to streamline event discovery
- Implemented a TypeScript backend with a Supabase database to handle user authentication and real-time event updates
- Implemented RESTful APIs and backend logic to manage asynchronous workflows while maintaining modular architecture

Sylvan Learning

Feb. 2025 – Aug. 2025

Instructor

Schaumburg, IL

- Tutored K–12 students in math, English, and SAT/ACT/AP test preparation using Sylvan's adaptive curriculum
- Led STEM courses for Sylvan's InZone summer camp at Harper College, delivering hands-on instruction in coursework including Lego Bot Labs, Coding Fundamentals, Python Programming, and STEM Challenges in Sustainable Energy

James B. Conant High School

Aug. 2024 – May 2025

Computer Science Teaching Assistant

Hoffman Estates, IL

- Earned EDU201 dual credit for assisting with a class of AP Computer Science A and tutoring all computer science students
- Developed the full curriculum for a new Arduino focused unit to be used in the following academic school years
- Debugged student code, graded projects and assessments, and created review guides to support student success

Revcor

Jun. 2024 - Aug. 2024

IT Technician

Carpentersville, IL

- Configured and deployed 80+ computer systems with updated software, resolving hardware-software integration issues
- Installed and upgraded internal CPUs, RAM, drives, and peripherals in existing company equipment to optimize workstations

PROJECTS

Sproutsy App Development

Jan. 2025 - May 2025

- Built a full-stack AI-integrated application using React and Firebase, incorporating external APIs and data-driven logic
- Engineered custom plant layout logic utilizing integrated geolocation, camera input, and weather tracking
- Created and presented a comprehensive marketing plan for the hypothetical release of the app at the end of development

Project Recycode

Jan. 2024 - May 2025

- Co-founded a local clothing donation organization that collected over 5,000 pounds of various clothing articles from libraries and schools, sorted them, and donated them to various centers in need such as homeless shelters and fabric recycling firms
- Designed the website and app for this organization using React Native, Supabase, Softr.IO, and multiple external API integrations

AI In Astrophysics Research Project

Jun. 2024 - Apr. 2025

- Built a machine learning model with 92% accuracy to predict exoplanet orbital periods using NASA stellar flux data
- Applied advanced techniques including regression analysis, data normalization, and feature engineering to maximize accuracy
- Authored a paper and delivered a formal presentation and seminar on the current applicable role of AI in astrophysics research

AWARDS

National Merit Scholarship Winner *presented by the National Merit Scholarship Corporation, 2025*

National Finalist in Cybersecurity and Digital Forensics *presented by Business Professionals of America, 2024*

National Qualifier in Fundamentals of HTML Web Design *presented by Business Professionals of America, 2025*

State Finalist in Python Programming *presented by Business Professionals of America, 2024*

CERTIFICATIONS & SKILLS

- **Certifications:** Certiport IT Specialist - HTML and CSS; CITI Research HIPAA
- **Technical Skills:** Python, Java, C/C++, TypeScript, JavaScript, SQL, PyTorch, TensorFlow, HTML, CSS, Linux, Git, LaTeX, Vim